

Problem Statement

Parkinson's disease is a progressive neurological disorder that primarily affects a person's movement, balance, and coordination. One of the most challenging symptoms is **Freeze of Gait (FoG)**, where patients suddenly become unable to move while walking. These unexpected freezing episodes often lead to falls, injuries, loss of confidence, and reduced independence in daily life.

Although several monitoring systems have been developed, many existing solutions focus on detecting symptoms only after they occur or require costly wearable devices and specialized medical equipment. Such limitations reduce their accessibility, especially for elderly patients and individuals living in remote or resource-limited areas.

The proposed project aims to develop an **AI-based Freeze of Gait Prediction and Smart Assistance System** that analyzes gait patterns using data collected from smartphones or wearable sensors. By identifying abnormal movement patterns before a freezing episode occurs, the system can provide real-time alerts through vibration, sound, or notifications, enabling patients to react quickly and reduce the risk of falls.

This solution combines Artificial Intelligence, Machine Learning, and healthcare technology to improve patient safety and quality of life. It is designed to be affordable, easy to use, and suitable for real-world deployment, while also assisting caregivers and healthcare professionals in monitoring patient conditions and supporting timely intervention.